separated by enormous distances, have temperatures that are so finely matched? A seemingly natural resolution to this puzzle is to note that, yes, two diametrically opposite places in the heavens are far apart today, but like twins separated at birth, during the earliest moments of the universe they (and everything else) were very close together. Since they emerged from a common starting point, you might suggest that it's not at all surprising that they share common physical traits such as their temperature.

In the standard big bang cosmology this suggestion fails. Here's why. A bowl of hot soup gradually cools to room temperature because it is in contact with the colder surrounding air. If you wait long enough, the temperature of the soup and the air will, through their mutual contact, become the same. But if the soup is in a thermos, of course, it retains its heat for much longer, since there is far less communication with the outside environment. This reflects that the homogenization of temperature between two bodies relies on their having prolonged and unimpaired communication. To test the suggestion that positions in space that are currently separated by vast distances share the same temperature because of their initial contact, we must therefore examine the efficacy of information exchange between them in the early universe. At first you might think that since the positions were closer together at earlier times, communication was ever easier. But spatial proximity is only one part of the story. The other part is temporal duration.

To examine this more fully, let's imagine studying a "film" of the cosmic expansion, but let's review it in reverse, running the film backward in time from today toward the moment of the big bang. Since the speed of light sets a limit to how fast any signal or information of any kind can travel, matter in two regions of space can exchange heat energy and thereby have a chance of coming to a common temperature only if the distance between them at a given moment is less than the distance light can have traveled since the time of the big bang. And so, as we roll the film backward in time we see that there is a competition between how close together our spatial regions become versus how far back we have to turn the clock for them to get there. For instance, if in order for the separation of our two spatial locations to be 186,000 miles, we have to run the film back to